# IGCSE 0607 Practice Questions — E1.17

## Surds

**Total Questions:** 33 — **Total Marks:** 161 Marks

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- **E1.17** Surds
  → **Paper 2 (Non-Calculator)** *(33 Qs)*


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**1** Simplify

$$\sqrt{75} - \sqrt{27} + \sqrt{12}$$

..... **[2]**

**2** Simplify.

$\sqrt{75} + \sqrt{48} - \sqrt{27}$

..... **[2]**

**3** Simplify fully.

$\sqrt{72} + \sqrt{50} - \sqrt{18}$

..... **[2]**

**4** Simplify

$$\sqrt{75} - \sqrt{27} + \sqrt{12}$$

..... **[2]**

**5** Simplify.

$\sqrt{75} + \sqrt{48} - \sqrt{27}$

..... **[2]**

**6** **(a)** Simplify.

$\sqrt{45}$

..... **[1]**

**(b)** Simplify.

$\sqrt{20} + \sqrt{80}$

..... **[2]**

**7** Simplify

$$\sqrt{75} + \sqrt{27} - \sqrt{12}$$

..... **[3]**

**8** Simplify the following expression, leaving your answer in the form $k\sqrt{3}$.

$\sqrt{48} - \sqrt{12}$

..... **[3]**

**9** **(a)** Simplify.

$\sqrt{75}$

..... **[2]**

**(b)** Simplify.

$2\sqrt{18} - \sqrt{8}$

..... **[2]**

**10** **(a)** **(i)** Simplify.

$$\sqrt{45} + \sqrt{20}$$

..... **[2]**

**(ii)** Rationalise the denominator.

$$\frac{18}{\sqrt{3}}$$

..... **[2]**

**11** **(a)** Simplify.

$\sqrt{45}$

..... **[1]**

**(b)** Work out the value of the following expression, giving your answer in the form $k\sqrt{2}$.

$\sqrt{72} + \sqrt{50} - \sqrt{18}$

..... **[2]**

**(c)** Rationalise the denominator.

$$\frac{21}{\sqrt{7}}$$

..... **[2]**

**12** **(a)** Expand and simplify.

$(4 - \sqrt{3})(2 + \sqrt{3})$

..... **[2]**

**(b)** Rationalise the denominator and simplify.

$$\frac{21}{\sqrt{7}}$$

..... **[2]**

**13** **(a)** Expand and simplify

$$(3 - \sqrt{5})^2$$

..... **[2]**

**(b)** Rationalise the denominator.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

**14** **(a)** Expand and simplify.

$$(3 - 2\sqrt{5})(4 + \sqrt{5})$$

..... **[2]**

**(b)** Rationalise the denominator and simplify.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

**15** **(a)** **(a)** Expand and simplify

$$(3 - 2\sqrt{5})(4 + \sqrt{5})$$

..... **[2]**

**(b)** Rationalise the denominator.

$$\frac{15}{\sqrt{3}}$$

..... **[2]**

**16** **(a)** Expand and simplify

$$(3+\sqrt{2})(4-\sqrt{2})$$

..... **[2]**

**(b)** Rationalise the denominator.

$$\frac{22}{5-\sqrt{3}}$$

..... **[3]**

**17** **(a)** Expand and simplify.

$$(2\sqrt{5}-3)(2\sqrt{5}+3)$$

..... **[2]**

**(b)** Rationalise the denominator.

$$\frac{14}{3-\sqrt{2}}$$

..... **[3]**

**18** **(a)** Expand and simplify.

$$(3\sqrt{2}-1)(2\sqrt{2}+5)$$

..... **[2]**

**(b)** Rationalise the denominator and simplify.

$$\frac{26}{4-\sqrt{3}}$$

..... **[3]**

**19** **(a)** Rationalise the denominator and simplify fully.

$$\frac{15}{\sqrt{5}}$$

..... **[2]**

**(b)** Expand and simplify.

$$(2+\sqrt{3})(5-\sqrt{3})$$

..... **[3]**

**20** **(a)** Expand and simplify.

$$(3+\sqrt{5})(2-\sqrt{5})$$

..... **[2]**

**(b)** Rationalise the denominator and simplify.

$$\frac{14}{\sqrt{7}}$$

..... **[3]**

**21** **(a)** Expand and simplify.

$$(5 - 2\sqrt{7})(3 + \sqrt{7})$$

..... **[2]**

**(b)** Rationalise the denominator and simplify fully.

$$\frac{22}{5 - \sqrt{3}}$$

..... **[3]**

**22** **(a)** Expand and simplify.

$(4 - 3\sqrt{2})(5 + 2\sqrt{2})$

..... **[2]**

**(b)** Rationalise the denominator and simplify.

$$\frac{11}{4 - \sqrt{5}}$$

..... **[3]**

**(c)** Solve the equation.

$x\sqrt{27} - 8 = x\sqrt{3} + 10$

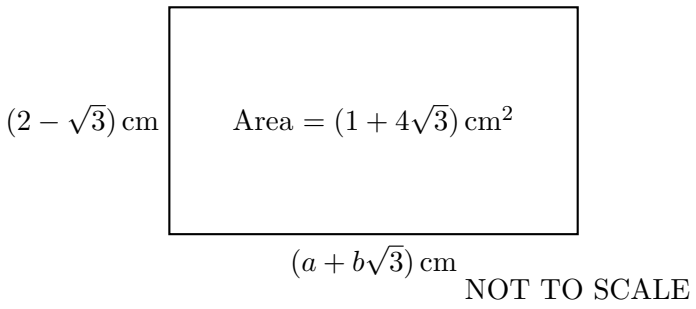
Give your answer in the form $k\sqrt{3}$ where $k$ is an integer.

$x =$ ..... **[3]**

**23** A rectangle has an area of $5 + \sqrt{7}$ cm$^2$ and a width of $3 - \sqrt{7}$ cm.
Calculate the length of the rectangle.
Give your answer in the form $a + b\sqrt{7}$ where $a$ and $b$ are integers.

..... *cm* **[4]**

**24** The diagram shows a rectangle with width $(2 - \sqrt{3})$ cm and length $(a + b\sqrt{3})$ cm, where $a$ and $b$ are integers.
The area of the rectangle is $(1 + 4\sqrt{3})\,\text{cm}^2$.




NOT TO SCALE

Work out the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[5]**

**25** **(a)** Rationalise the denominator and simplify.

$$\frac{5+\sqrt{3}}{2-\sqrt{3}}$$

..... **[3]**

**(b)** Solve the equation.
Give your answer in the form $a+b\sqrt{2}$, where $a$ and $b$ are integers.

$x(3-\sqrt{2})=14$

$x =$ ..... **[3]**

**26** **(a)** Express $\dfrac{5+\sqrt{3}}{2-\sqrt{3}}$ in the form $a + b\sqrt{3}$, where $a$ and $b$ are integers.

..... **[3]**

**(b)** A rectangle has an area of $12 + 4\sqrt{5}$ square centimetres and a width of $3 + \sqrt{5}$ centimetres. Calculate the exact length of the rectangle, giving your answer in its simplest form.

Length = ..... cm **[4]**

**27** **(a)** Rationalise the denominator and simplify.

$$\frac{22}{5-\sqrt{3}}$$

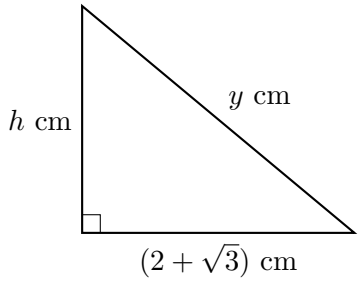
..... **[3]**

**(b)** Solve the equation.

$x\sqrt{2} = x + \sqrt{8}$ Give your answer in the form $a + b\sqrt{2}$, where $a$ and $b$ are integers.

$x =$ ..... **[4]**

**28** A right-angled triangle has a base of $(2 + \sqrt{3})$ cm and an area of $(5 + 3\sqrt{3})$ cm$^2$.




**(a)** Show that the exact height, $h$, is $(2 + 2\sqrt{3})$ cm.

**[4]**

**(b)** The hypotenuse of the triangle is $y$ cm.
Work out the exact value of $y^2$.
Give your answer in the form $a + b\sqrt{3}$, where $a$ and $b$ are integers.

$y^2 =$ ..... **[3]**

**29** **(a)** A rectangle has an area of $(5 + 2\sqrt{3})$ cm$^2$ and a length of $(2 + \sqrt{3})$ cm.

Work out the width of the rectangle in the form $a + b\sqrt{3}$, where $a$ and $b$ are integers.

..... cm **[3]**

**(b)** Solve the equation.

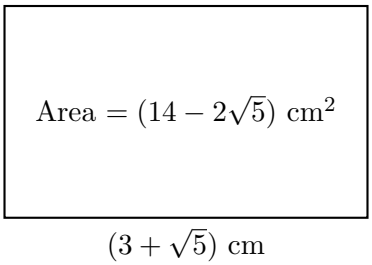
$$x\sqrt{5} - 2 = 2x + \sqrt{20}$$

Give your answer in the form $c + d\sqrt{5}$, where $c$ and $d$ are integers.

Show all your working.

$x =$ ..... **[4]**

**30** A rectangle has length $(3 + \sqrt{5})$ cm and area $(14 - 2\sqrt{5})$ cm$^2$.




**(a)** Calculate the width of the rectangle.
Give your answer in the form $a + b\sqrt{5}$, where $a$ and $b$ are integers.

..... cm **[4]**

**(b)** The length of the diagonal of the rectangle is $d$ cm.
Show that $d^2 = 308 - 124\sqrt{5}$.

**[3]**

**31** **(a)** Rationalise the denominator and simplify fully.

$$\frac{6}{3-\sqrt{3}}$$

..... **[4]**

**(b)** Write the following expression in the form $a\sqrt{2}+b\sqrt{3}$, where $a$ and $b$ are integers.

$$\frac{\sqrt{24}+\sqrt{54}}{\sqrt{3}}-\frac{10}{\sqrt{2}}$$

..... **[3]**

**32** A right-angled triangle has a base of length $(3+\sqrt{2})$ cm and an area of $(5+2\sqrt{2})$ cm$^2$.

**(a)** Show that the height of the triangle is $\dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ cm.

**[2]**

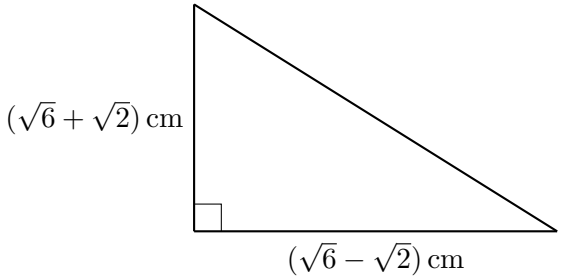
**(b)** Rationalise the denominator of $\dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ to express the height in the form $\dfrac{a+b\sqrt{2}}{c}$, where $a$, $b$ and $c$ are integers.

..... cm **[4]**

**(c)** Given instead that another expression for a length is $\dfrac{1}{\sqrt{7}-2}$, write this length in the form $\dfrac{\sqrt{7}+c}{d}$ where $c$ and $d$ are integers.

..... **[2]**

**33** The diagram shows a right-angled triangle.




**(a)** Calculate the area of the triangle.

..... $cm^2$ **[3]**

**(b)** Show that the length of the hypotenuse is 4 cm.

**[4]**

**(c)** Another right-angled triangle is mathematically similar to this triangle.
The area of the similar triangle is 18 $cm^2$.
Calculate the length of its hypotenuse.

..... cm **[3]**